CFPB Complaint Intelligence System

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Abstract

This report presents an end-to-end complaint intelligence system for CFPB complaint narratives. The system integrates multiclass complaint classification, binary high-risk prediction, calibrated probability scoring, explainability outputs, and triage-ready exports. The objective is to transform unstructured complaint data into actionable risk signals for regulatory prioritization.

The framework emphasizes not only predictive performance, but also interpretability, calibration, and operational usability. The final system enables ranking, explanation, and structured triage of complaints, making it suitable for real-world regulatory and financial decision-support workflows.

1 Introduction

Consumer complaint operations require practical tools for organizing large volumes of unstructured narrative text into actionable queues. In financial services and regulatory environments, classification alone is insufficient. Institutions must also determine which complaints require urgent attention and understand the reasoning behind such prioritization.

This project develops a unified complaint intelligence system that:

- classifies complaints into product categories,
- predicts high-risk complaints,
- generates calibrated risk scores,
- provides explainable outputs, and
- produces triage-ready ranked outputs.

2 Business Problem

The central question addressed is:

> Which complaints should be reviewed first, and why?

This requires combining prediction, prioritization, and interpretability into a single framework.

3 Dataset and Targets

3.1 Multiclass Target

Complaint **product** classification:

- Checking or savings account
- Credit card
- Credit reporting
- Debt collection
- Mortgage

3.2 Binary Target

`high_risk` defined as:

- Disputed complaint
- Untimely response
- Vulnerable consumer tag

4 Methodology

4.1 Models

Multiclass:

- TF-IDF + Logistic Regression

Binary Risk:

- Baseline: Logistic Regression
- Advanced: Calibrated Linear SVM

4.2 Calibrated Risk Score

$$\hat{p}(Y = 1 \mid X)$$

This probability is used directly for ranking and triage.

4.3 Risk Interpretation

- Low risk: $0.0 - 0.4$
- Medium risk: $0.4 - 0.7$
- High risk: $0.7 - 1.0$

5 Results

5.1 Multiclass Performance

- Accuracy: 0.8996
- Macro F1: 0.7829

5.2 Confusion Matrix

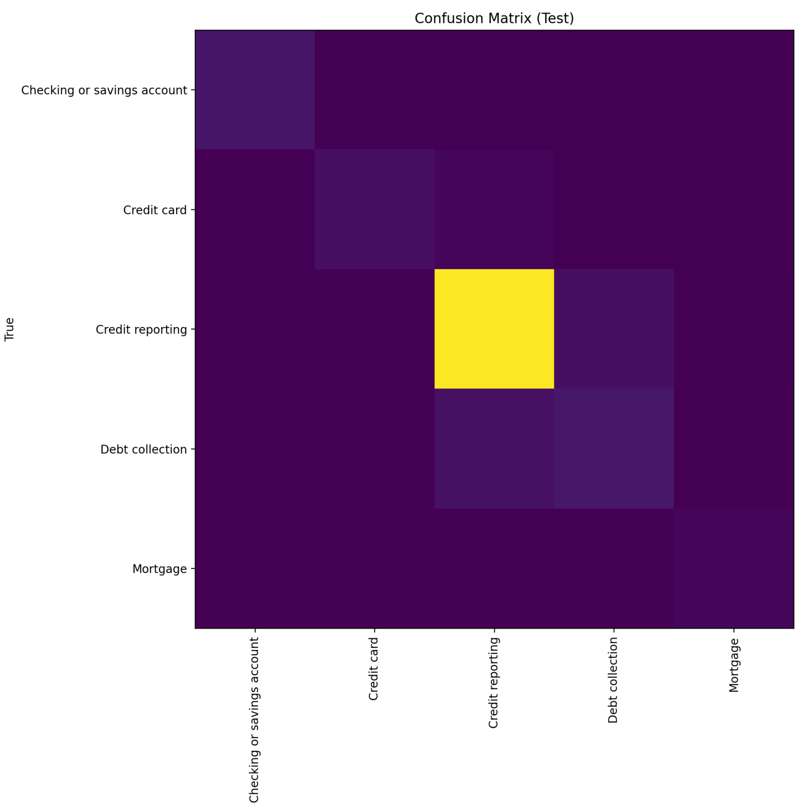

Figure 1: Multiclass classification confusion matrix

Interpretation The model performs strongly on the dominant class (Credit reporting), while showing moderate confusion in smaller categories such as Debt collection, indicating class imbalance and overlapping semantic structures.

5.3 Binary Risk Models

Model	ROC-AUC	Avg Precision	Brier Score
Logistic Regression	0.7275	0.1917	0.1146
Calibrated SVM	0.6328	0.1482	0.0765

Table 1: Binary risk model comparison

Interpretation The logistic model demonstrates stronger ranking capability, while the calibrated SVM produces more reliable probability estimates. This highlights the trade-off between ranking performance and calibration quality in risk modeling.

5.4 ROC Curve

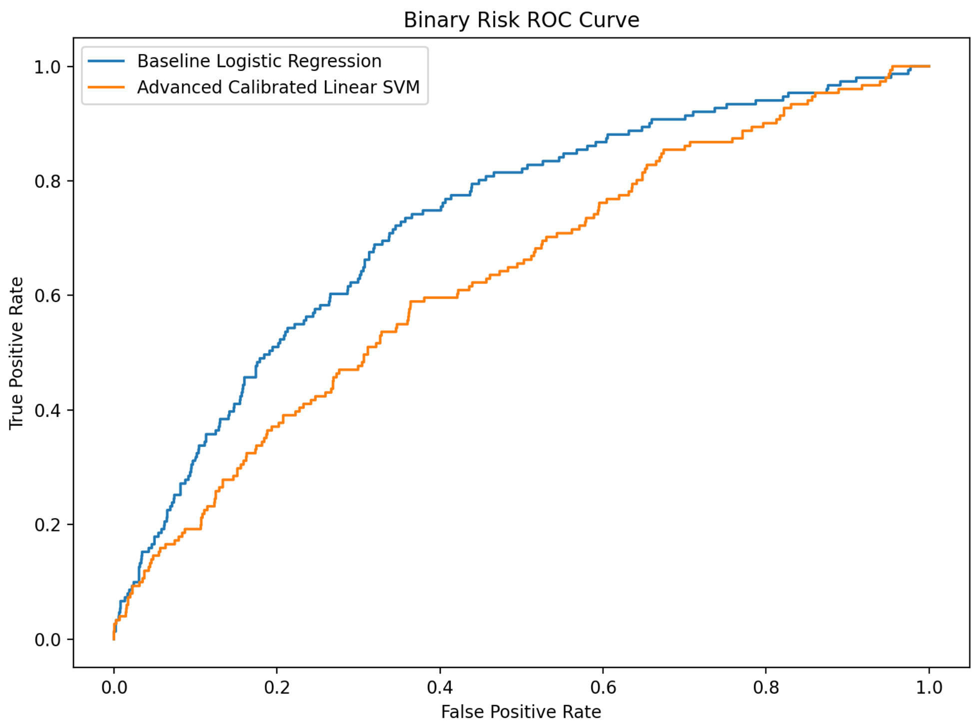

Figure 2: ROC curve comparison

5.5 Precision-Recall Curve

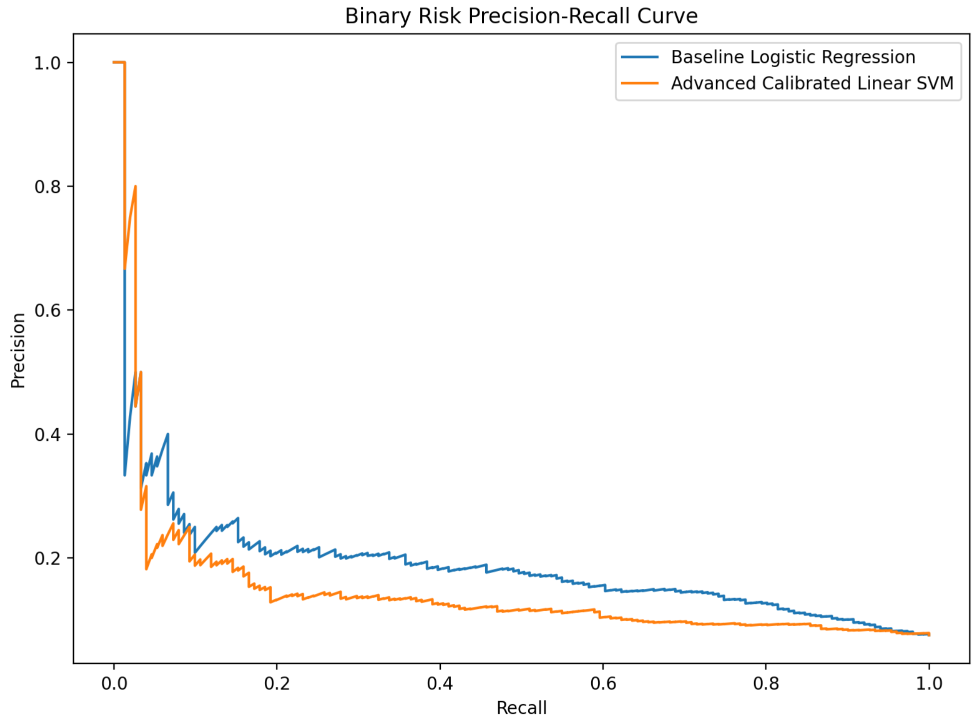

Figure 3: Precision-recall curve

5.6 Calibration Curve

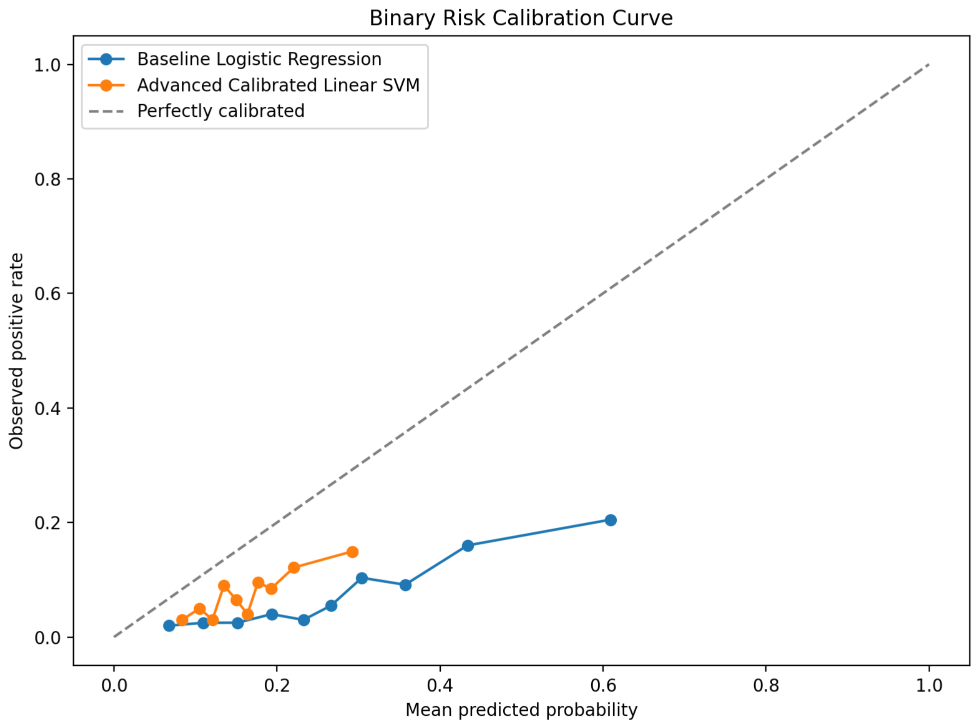

Figure 4: Calibration curve

Interpretation Calibration analysis demonstrates that probability estimates differ significantly across models. For triage systems, calibrated probabilities are critical because they directly influence prioritization decisions.

6 Explainability

The system combines:

- Target-based signals
- Rule-based signals
- Model-based feature contributions

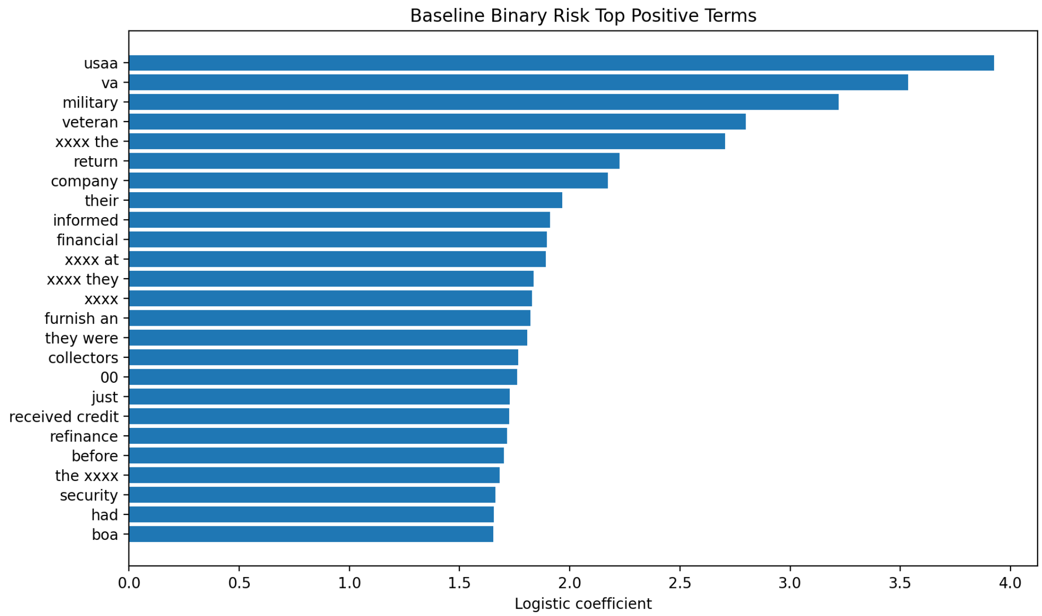

Figure 5: Top contributing terms for high-risk classification

Interpretation The model identifies institutional, dispute-related, and consumer-protection keywords as strong indicators of elevated complaint risk.

7 Triage Output

The system produces:

- Ranked complaints
- Risk scores
- Explanation strings

This enables real-world workflows where analysts prioritize cases based on both severity and interpretability.

8 Discussion

A key insight is that complaint intelligence systems must balance predictive performance with operational usability.

While ranking-based models maximize detection, calibrated models provide more reliable decision thresholds. This trade-off is central in risk systems where interpretability and stability are required.

9 Conclusion

This study demonstrates that integrating classification, calibration, and explainability creates a robust framework for complaint intelligence.

The system highlights that effective decision-support models are not defined solely by predictive accuracy, but by their ability to provide interpretable, calibrated, and actionable outputs in real-world workflows.

Future work includes:

- transformer-based models
- improved labeling strategies
- real-time deployment pipelines